

Bash Language Summary

1. Introduction

Bash (Bourne Again Shell) is a command language interpreter for Unix/Linux systems. It's used for executing commands, automating tasks, and writing shell scripts.

2. Basic Commands

- ls – list directory contents
- cd – change directory
- pwd – print working directory
- mkdir – make directory
- rm – remove files or directories
- cp – copy files
- mv – move or rename files
- cat – view file content
- echo – print text or variables
- man – show manual pages

3. Variables

Variables store data temporarily. Example:

name='Uthman'

To access: **\$name**

4. Conditionals

Bash supports if-else statements:

```
if [ $age -ge 18 ]; then
    echo 'Adult'
else
    echo 'Minor'
fi
```

5. Loops

for loops iterate over lists:

```
for i in {1..5}
do
  echo $i
done
```

while loops run until a condition is false:

```
while [ $x -lt 10 ]
do
  echo $x
  ((x++))
done
```

6. Functions

Functions group reusable code:

```
greet() {
  echo 'Hello, $1!'
}
```

Call with: **greet Uthman**

7. File Permissions

Use **chmod** to change file permissions.

Examples:

- **chmod +x script.sh** → make executable
- **chmod 755 file** → set read/write/execute for owner

8. Redirection & Pipes

Redirect output or connect commands:

- **>** overwrite file
- **>>** append to file
- **<** input from file
- **|** pipe output to another command

9. Script Execution

Make script executable and run it:

- **chmod +x script.sh**

- ./script.sh

End of Summary